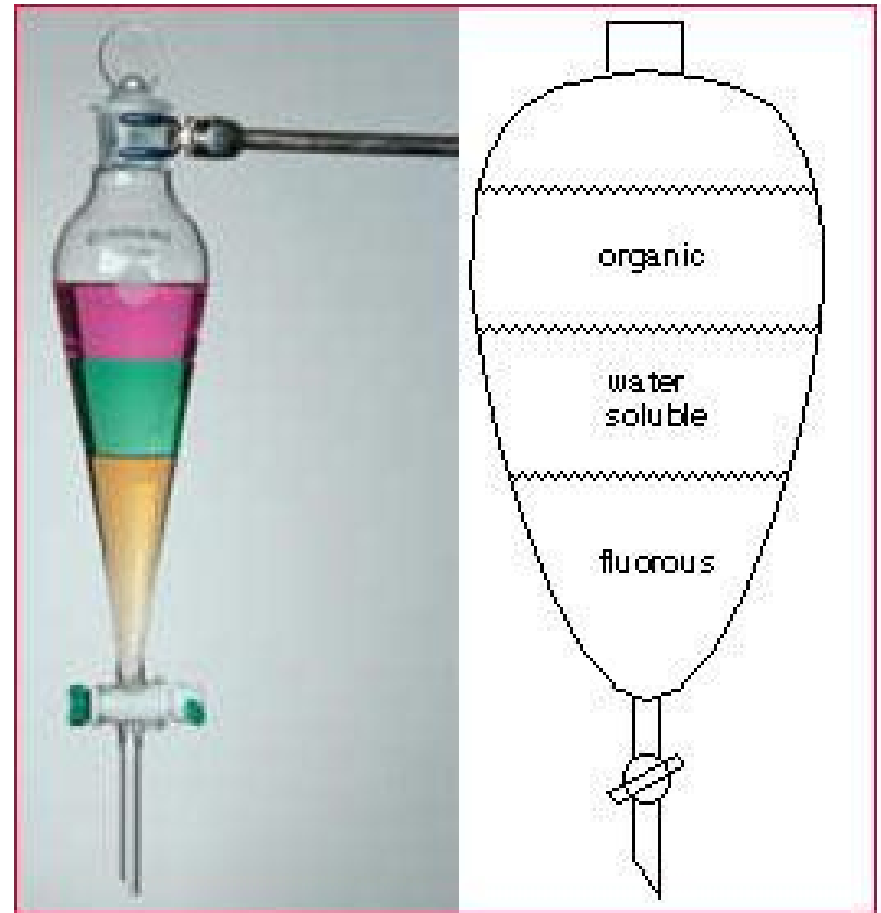


Phase equilibria and the phase rule

Phase equilibria and the phase rule

- **Definition:**
 - ✓ **Phase** is a homogeneous, physically distinct portion of a system that is separated from other portions of the system by bounding surfaces.



Phase equilibria and the phase rule

- The three primary phases (solid, liquid, and gas) of matter are defined individually under different conditions, but in most systems we encounter phases in coexistence.
- Examples of phases coexistence:
 - ✓ A mixture of ice and water has **2 phases** which are solid and liquid.

Phase equilibria and the phase rule

- ✓ A mixture of oxygen gas and nitrogen gas has **1 phase** which is gas phase (the system is homogeneous).
- ✓ A glass of ice water comprising: ice, water, and water vapour has **3 phases**.

Phase equilibria and the phase rule

- ✓ A mixture of oil and water has **2 phases** (liquid). Oil and water are not mixable and have the boundaries to separate both phases.
- Phases coexistence can only occur over a limited range.
 - ✓ For example, ice does not last as long in boiling water as it does in cold water.

The phase rule

- J. Willard Gibbs formulated the phase rule, which is a relationship for determining the least number of intensive variables (independent variables) that do not depend on the size of the phase, e.g., **temperature, pressure, concentration, etc**) that can be changed without changing the equilibrium state of the system.

The phase rule (cont'd)

- The phase rule: $F = C - P + 2$
- **F, degree of freedom**, is the least number of intensive variables (e.g., temperature, pressure, concentration, etc) that can be changed without changing the equilibrium state of the system.
- **F** is the number of variables that can be changed **independently** without causing appearance of a new phase or disappearance of an existing phase.

The phase rule (cont'd)

➤ **C: number of components**

- ✓ the number of constituents that can be expressed in the form of chemical formula.

➤ **P: Number of phases**

The phase rule (cont'd)

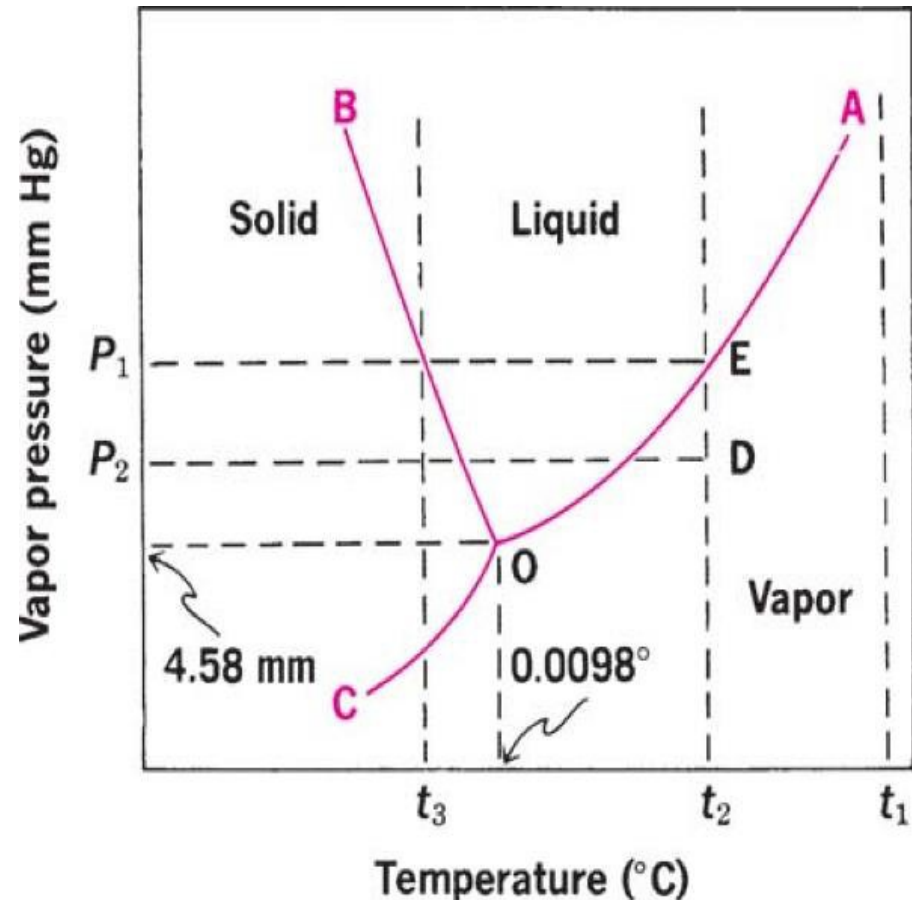
- Example,
 - ✓ the number of components (**C**) in the equilibrium mixture of ice, liquid water and water vapor = 1 because the composition of all the three phases is described by the chemical formula H_2O .

System containing one component: e.g., water

- Phase diagram is graphical representation which indicates the phase equilibrium.
- Each area correspond to a single phase.
- **The number of degrees of freedom in each area** is : $F = C - P + 2$

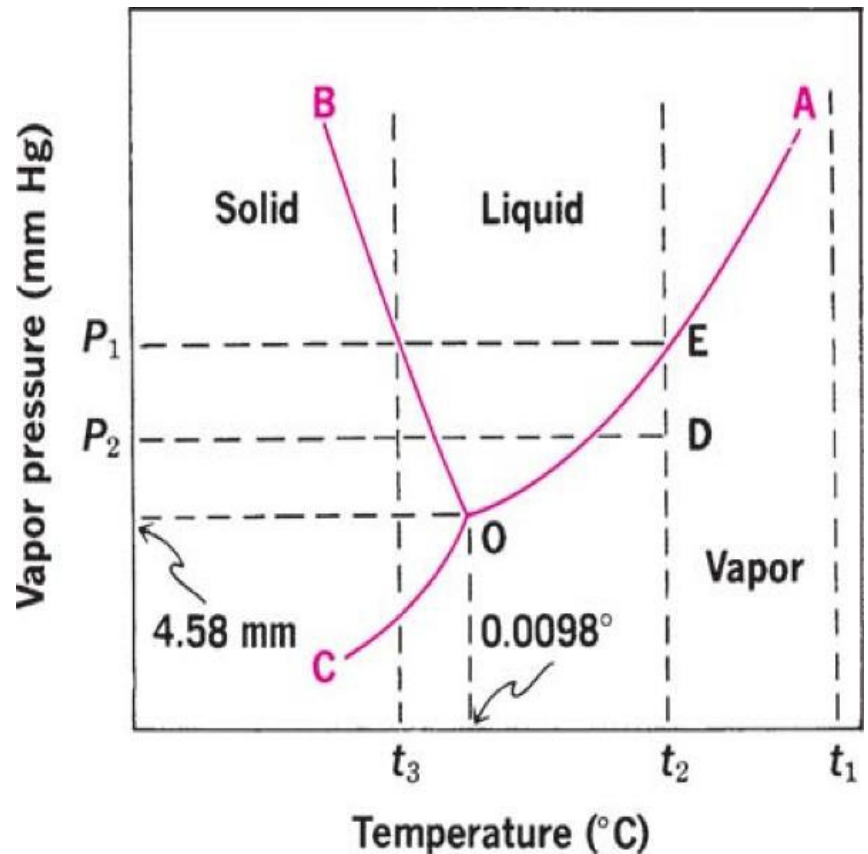
$$F = 1 - 1 + 2 = 2$$

- Fig. Phase diagram of water at moderate temperatures:



System containing one component: e.g., water

- $F = 2$ for phase diagram of water means that **temperature and pressure** can be varied independently within these areas without change in number of phases.
- **$F = 2$ (bivariant system).**
- Two variables can be changed independently without losing the phase

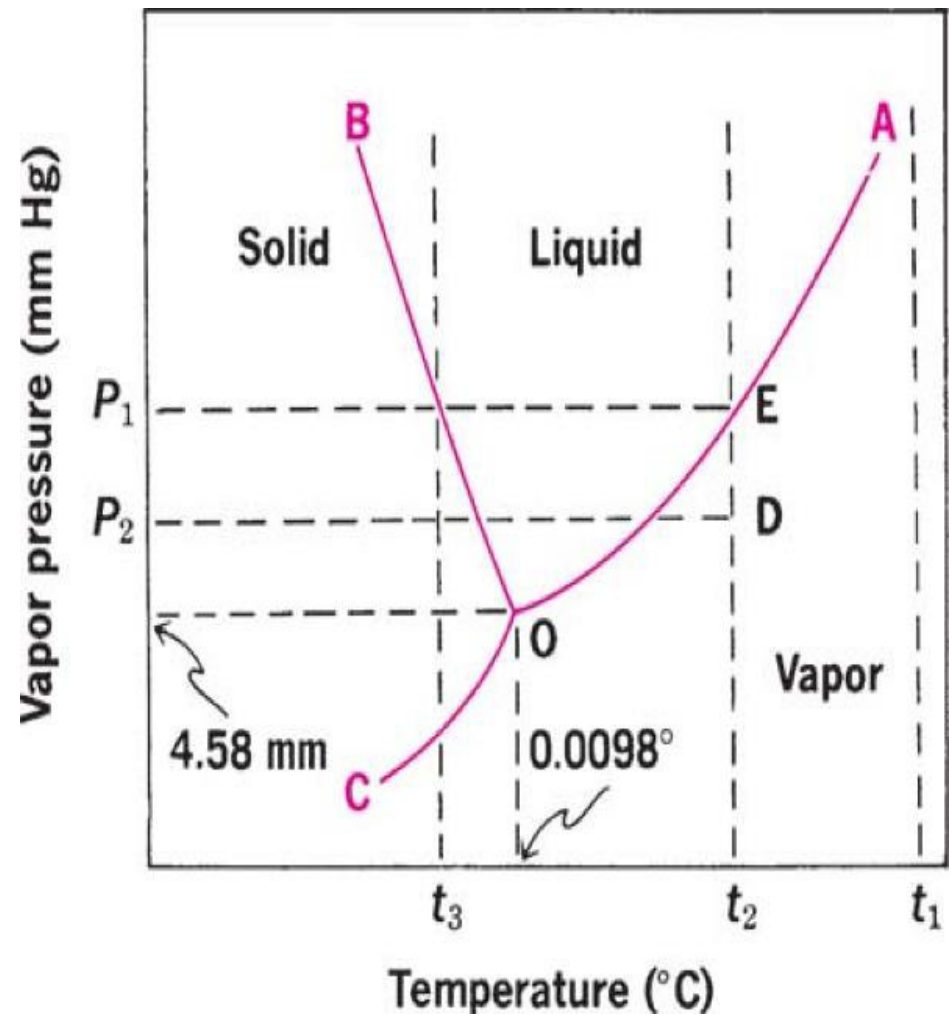


System containing one component: e.g., water

□ Equilibrium boundaries:

- For points that lie on one of the lines **AO**, **BO**, or **CO**, these lines form the equilibrium boundaries between 2 phases.

- $F = 1 - 2 + 2 = 1$



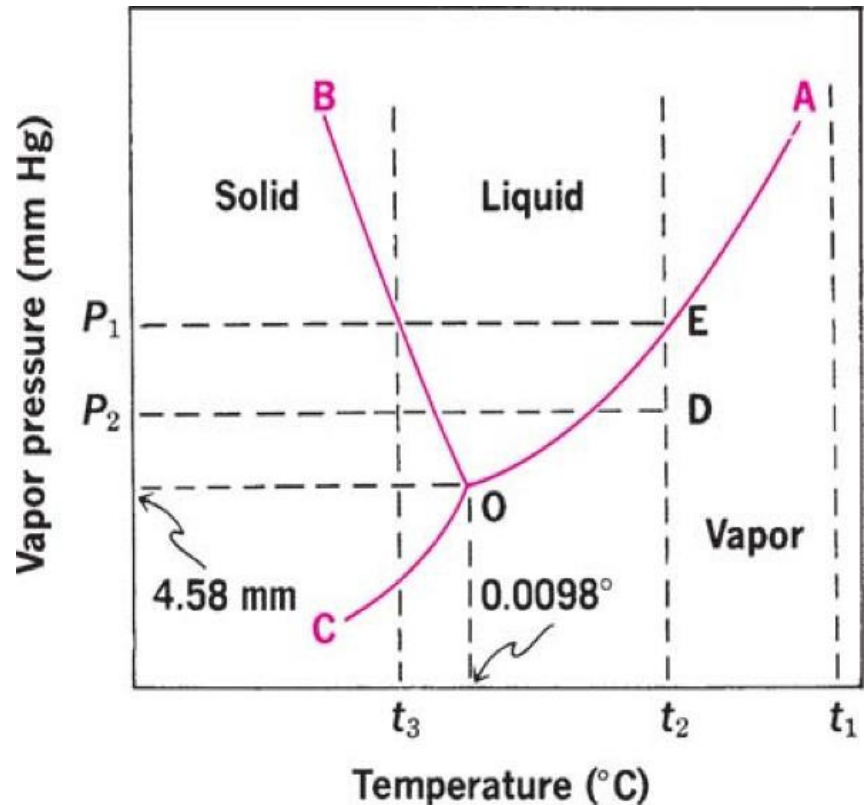
System containing one component: e.g., water

- $F=1$ is a **univariant** system.
- ➔ Only one independent variable should be altered at a time for the two phases to remain at equilibrium.
- ➔ If two or more independent variables are changed, one of the phases disappears.

System containing one component: e.g., water

■ Equilibrium boundaries represents:

- ✓ melting point/Freezing point
- ✓ Vaporization point/Condensation point
- ✓ Sublimation point/Deposition point

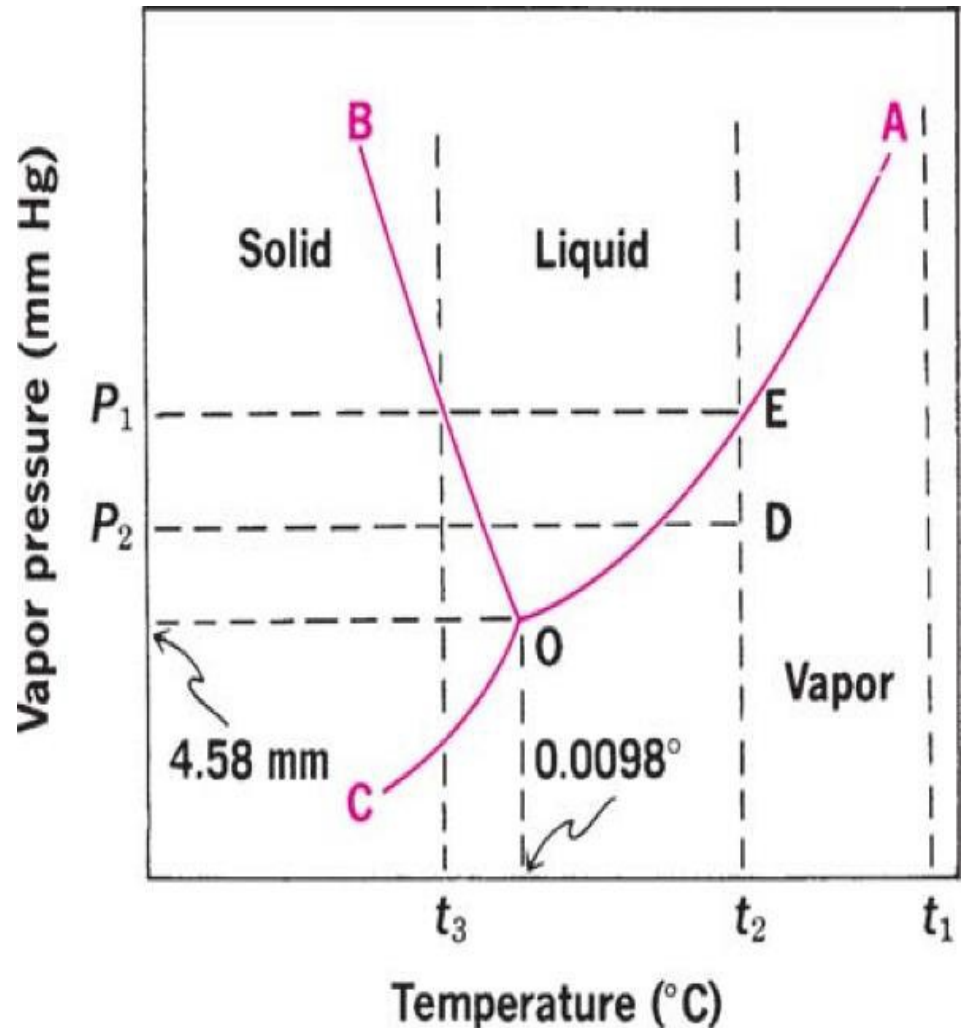


Triple point

- The boundary lines meet at **O**, which is the only point in the diagram where 3 phases may coexist in equilibrium and it is therefore termed a triple point.

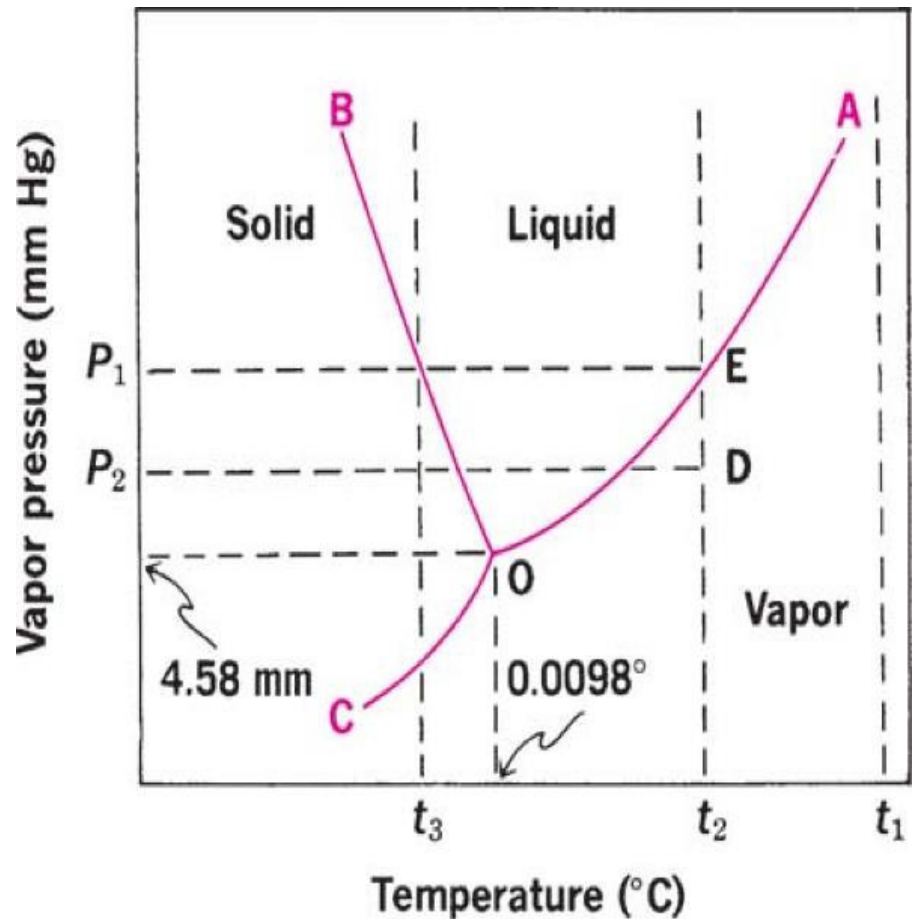
✓ $T = 0.0098\text{ }^{\circ}\text{C}$

✓ $P = 4.58\text{ mmHg}$



Triple point

- At triple point,
$$F = 1 - 3 + 2 = 0$$
- The system is **invariant**; i.e., any change in P or T will result in an alteration of the number of phases that are at equilibrium.



Triple point.....

- Suppose we cool liquid water and its vapor until a third phase (ice) separates out. We get three phases (ice-water-water vapour).
- The three phases exist at equilibrium only at $T = 0.0098\text{ }^{\circ}\text{C}$ and $P = 4.58\text{ mmHg}$. But not on any other T or P .
- If we attempt to vary the particular conditions (i.e., temperature or pressure) necessary to maintain this system, we will lose a phase.

The phase rule (cont'd)

- As the number of components increases, so do the number of degree of freedom required to define the system.
- The greater the number of phases in equilibrium, however, the fewer are the degrees of freedom.

Phase equilibria of two-component system

- A maximum of three degrees of freedom is possible in a two-component system, for example, temperature, pressure and concentration.

$$F = C - P + 2 \Rightarrow 2 - 1 + 2 = 3$$

Phase equilibria of two- component system

- Because in practice we are primarily concerned with liquid and/or solid phases in the particular system under examination, we frequently choose to disregard the vapor phase and work under normal conditions of 1 atm (760 mm Hg) of pressure.
 - In this manner we reduce the number of degrees of freedom by one.

Phase equilibria of two- component system (cont'd)

- In a two-component system, therefore, only two variables (e.g., temperature and concentration) remain, and we are able to portray the interaction of these variables by the use of planer figures on a rectangular–coordinate graph paper.

Phase equilibria of two- component system (cont'd)

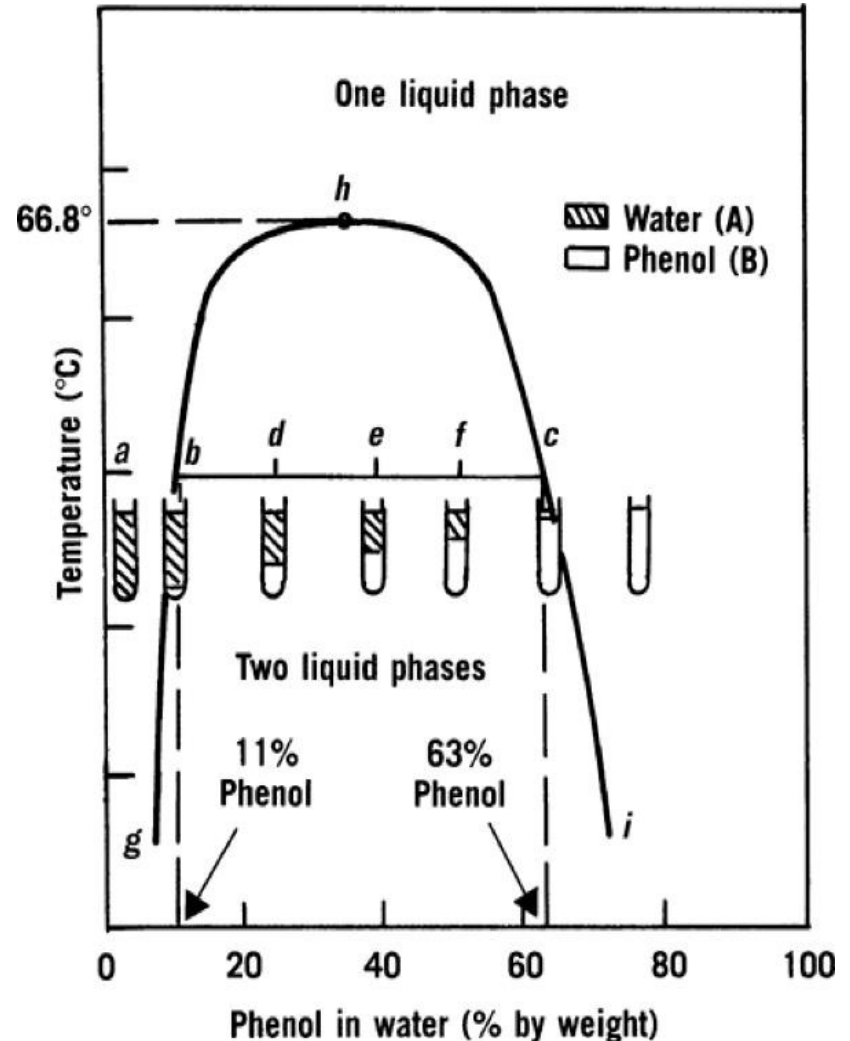
- Systems in which the vapor phase is ignored and only solid and/or liquid phases are considered are termed *condensed systems*.
- Condensed systems are most appropriate for solid and liquid dosage forms.

Two-component systems containing liquid phases

- Water and ethyl alcohol are miscible in all proportions.
- Water and mercury are, for all practical purposes, completely immiscible regardless of the relative amounts of each present.
- Between these extremes lies a whole range of systems that exhibit partial miscibility.
e.g., phenol in water

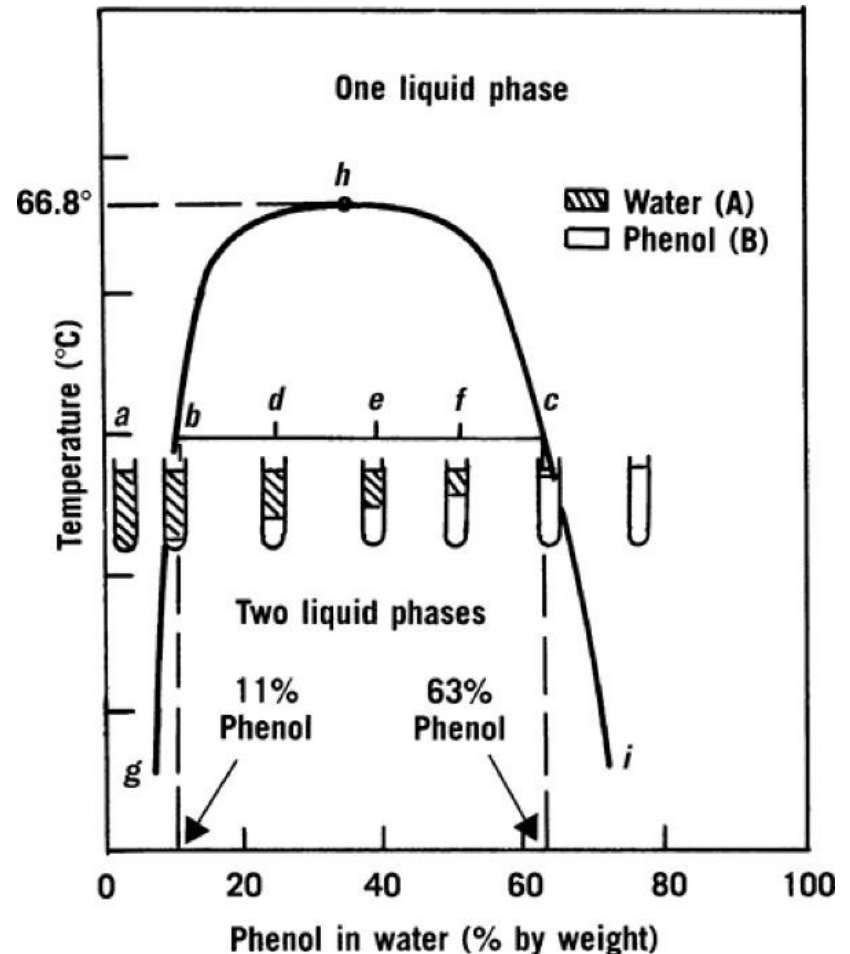
Two-component systems containing liquid phases

- Phenol and water exhibit partial miscibility (or partial immiscibility).
- Curve *gbhci* shows the limits of temperature and concentration within which two liquid phases exist in equilibrium.



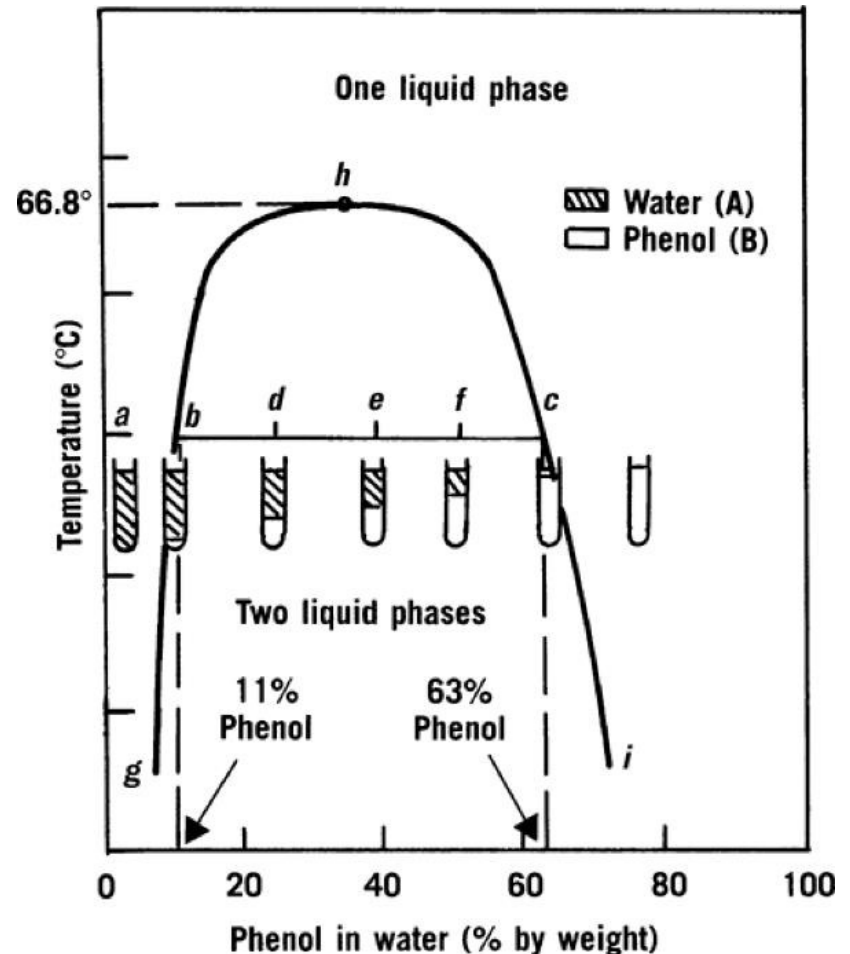
Two-component systems containing liquid phases

- The region outside this curve contains systems having one liquid phase.



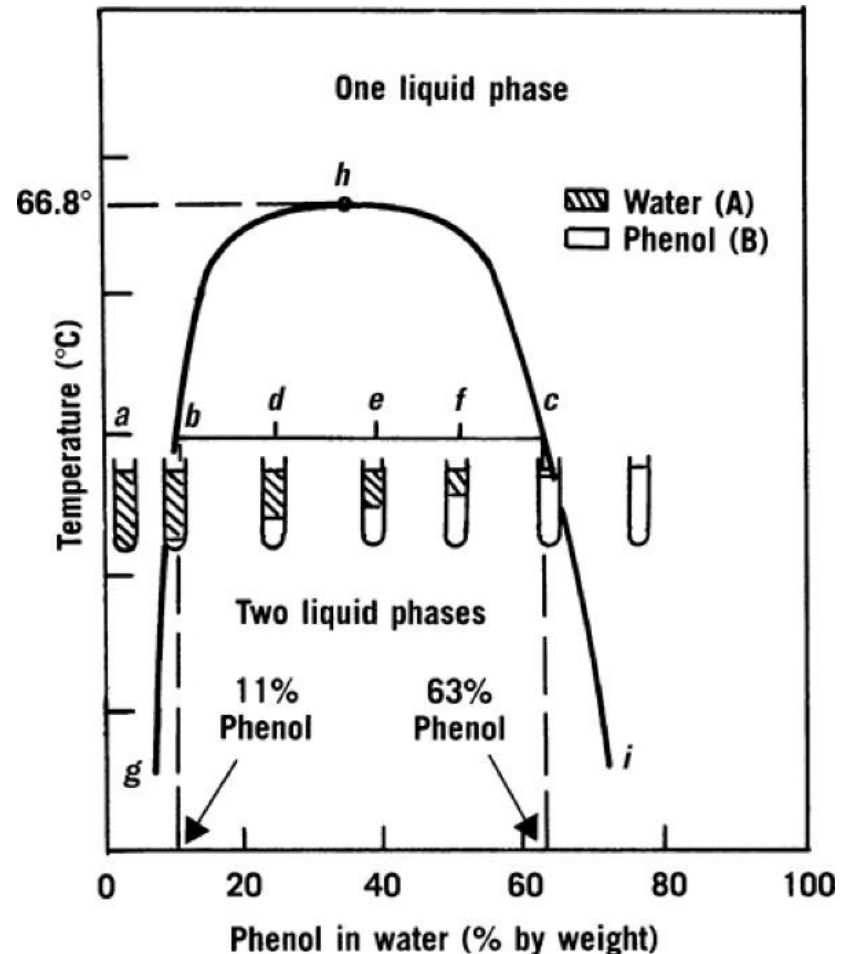
Two-component systems containing liquid phases

- Starting at the point a, equivalent to a system containing 100% water (i.e., pure water) at 50°C, adding known increments of phenol, the whole being maintained at 50°C, will result in the formation of a single liquid phase until the point b is reached, at which point a minute amount of a second phase appears.



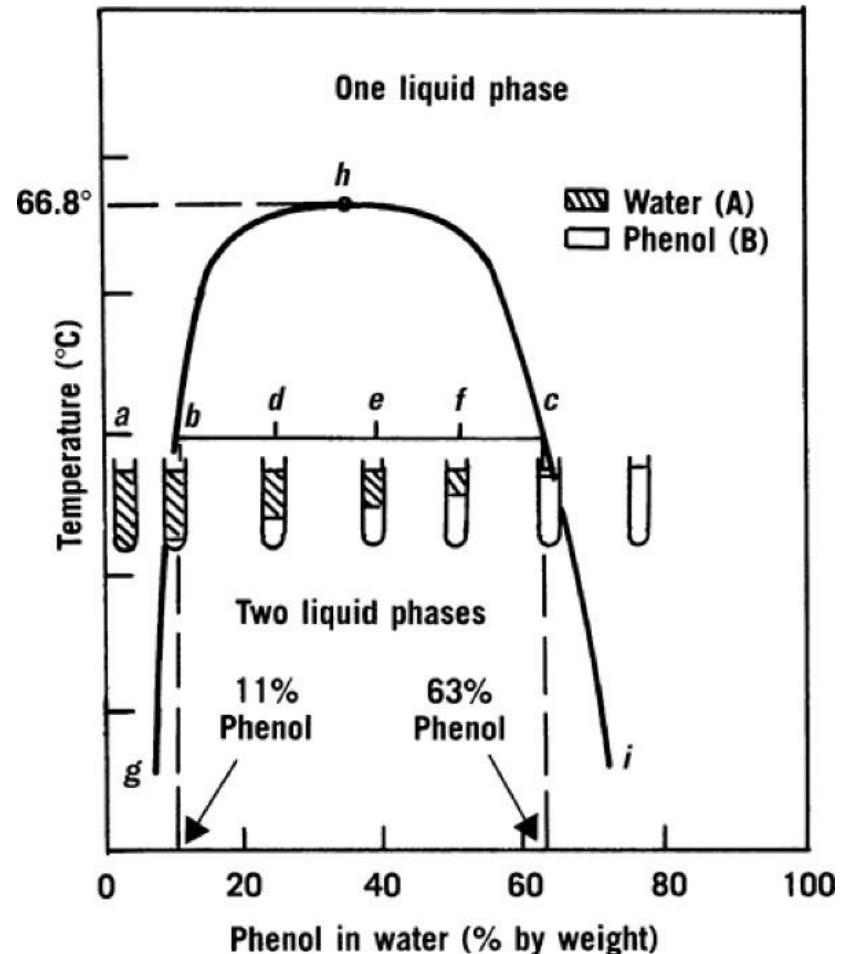
Two-component systems containing liquid phases

- The concentration of phenol at which this occurs is 11% by weight of phenol in water.
- Analysis of the second phase, which separates out on the bottom, shows it to contain 63% by weight of phenol in water (as point **C**).



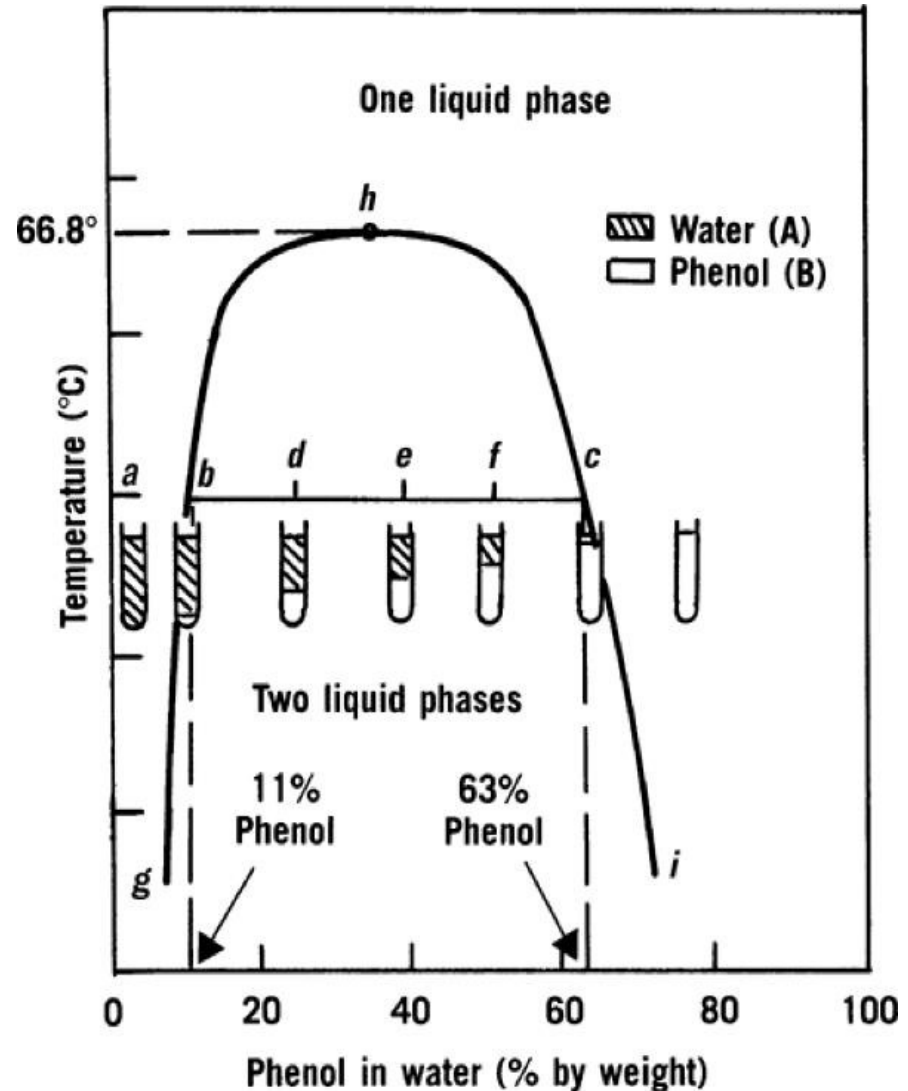
Two-component systems containing liquid phases

- Mixtures (**b** to **c**) are containing increasing amounts of phenol.
- Thus, b to C are systems in which the amount of the phenol-rich phase (B) continually increases and the amount of the water-rich phase (A) decreases.



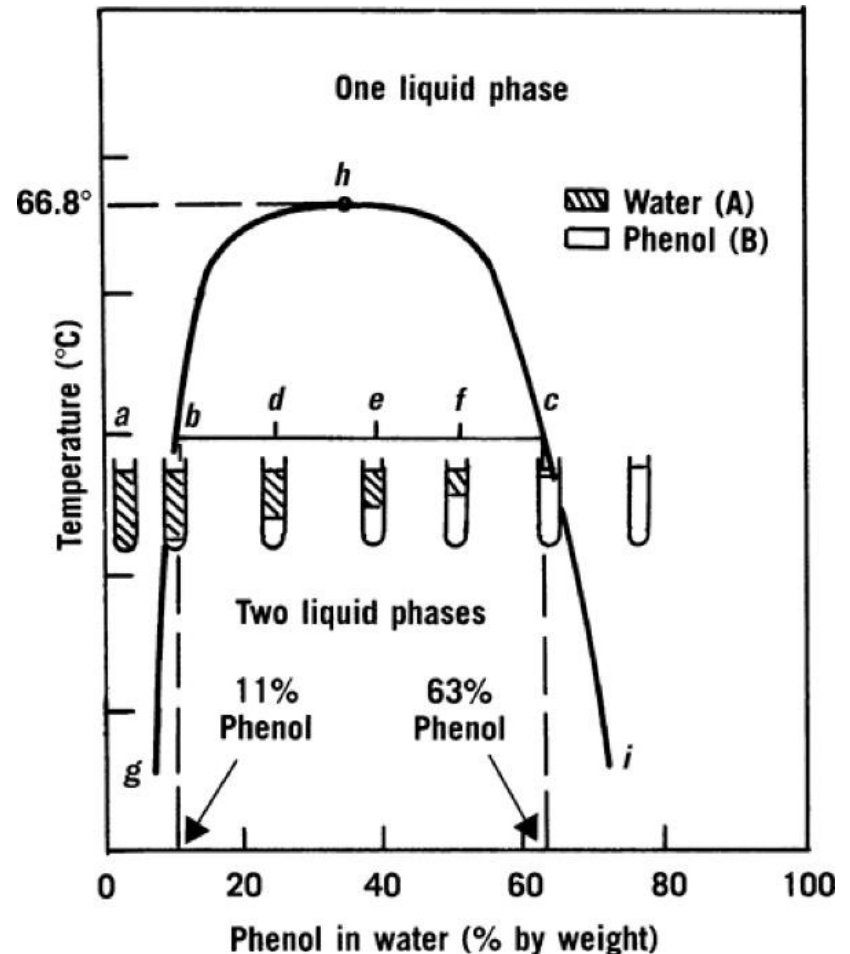
Two-component systems containing liquid phases

- Once the total concentration of phenol exceeds 63% at 50°C, a single phenol-rich liquid phase is formed.
- The maximum temperature at which the two-phase region exists is termed *the critical solution, or upper consolute temperature*.



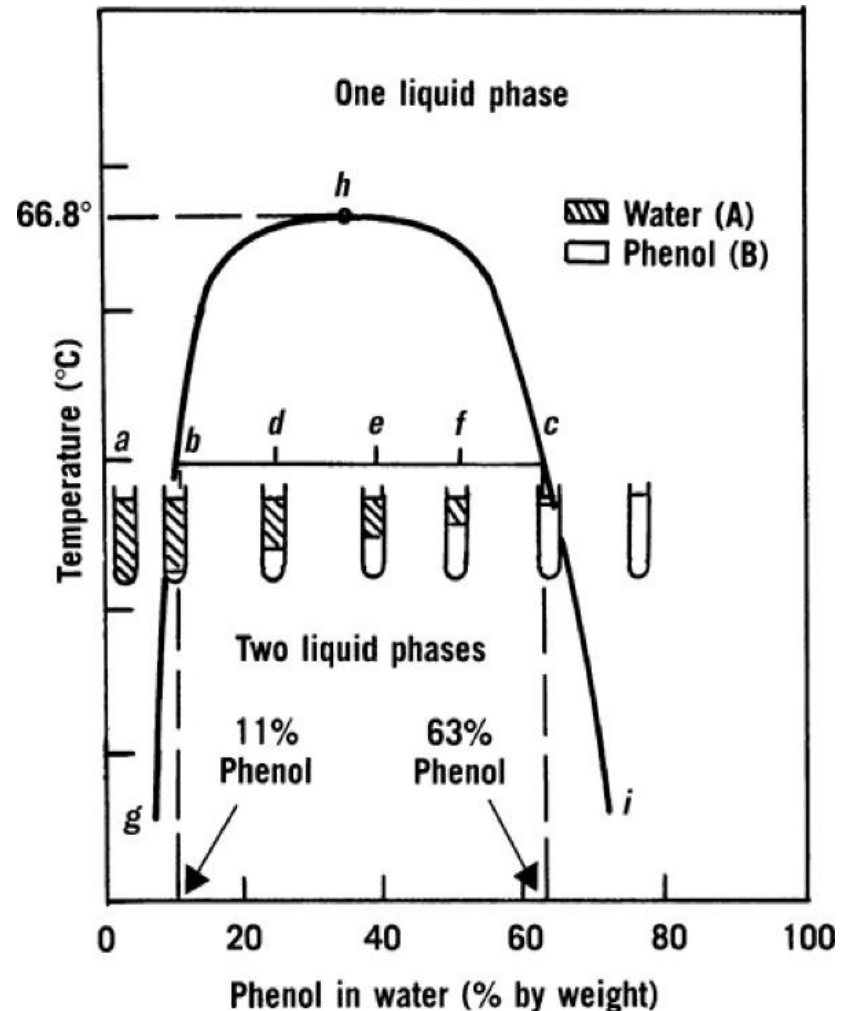
Two-component systems containing liquid phases

- In the case of phenol water system, this is 66.8 °C.
- All combinations of phenol and water above this temperature are completely miscible and yield one-phase liquid systems.



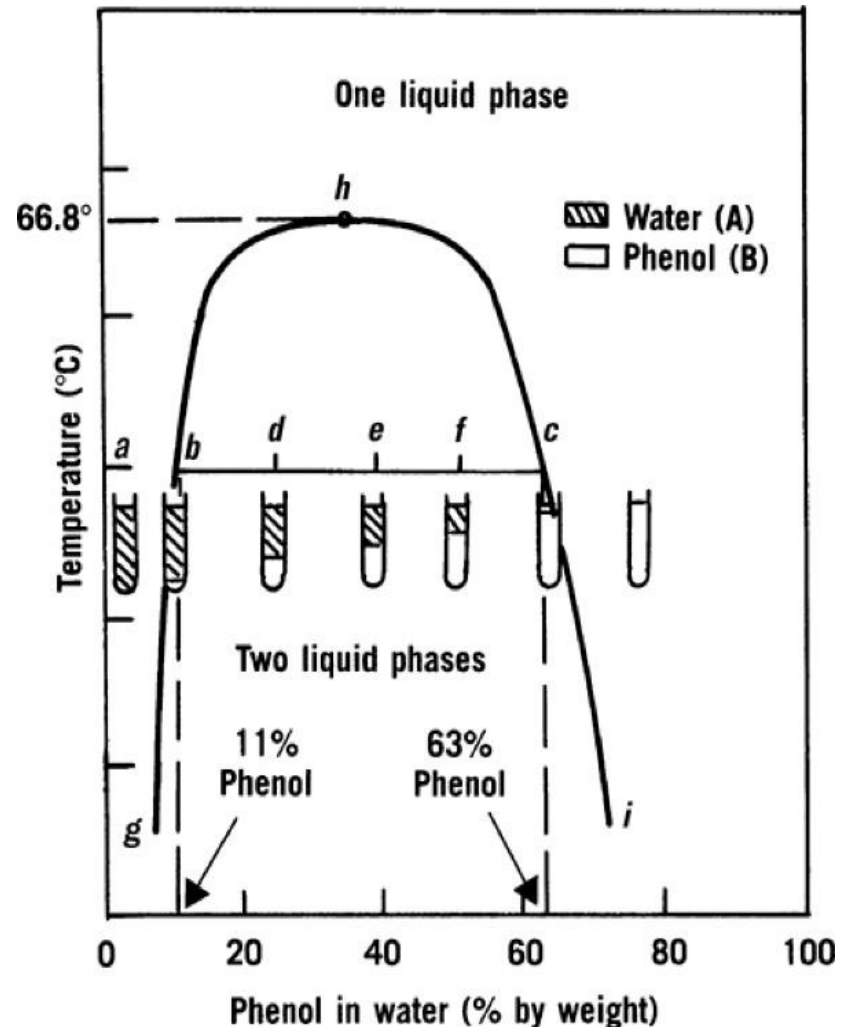
Two-component systems containing liquid phases

- The line *bc* drawn across the region containing two phases is termed a *tie line*; it is always parallel to the base line in two component systems.
- *All systems prepared on a tie line, at equilibrium, will separate into phases of constant composition.*
- These phases are termed *conjugate phases*.



Two-component systems containing liquid phases

- For example, any system represented by a point on the line *bc* at 50°C separates to give a pair of conjugate phases whose compositions are *b* and *c*.
- However, the relative amounts of the two phases vary.

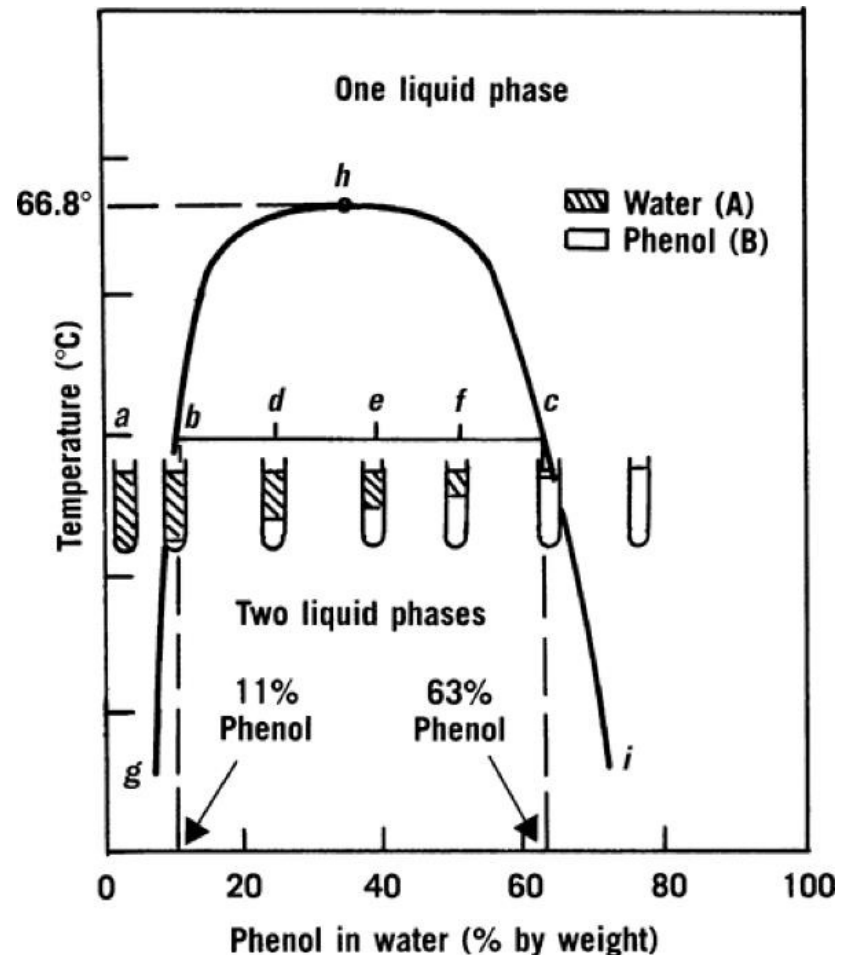


Two-component systems containing liquid phases

- Relative amounts of phenol and water phases of the two liquid layers are then calculated by the following formula.

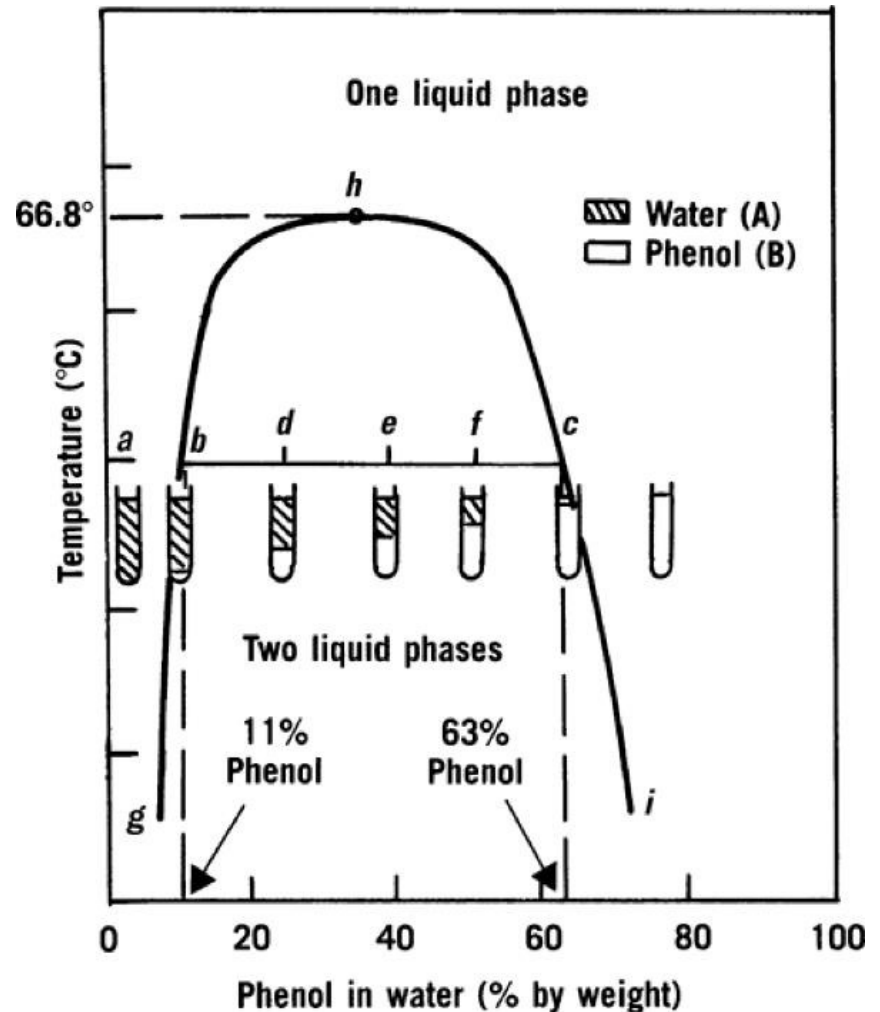
$$\frac{\text{Weight of Phase A}}{\text{Weight of Phase B}} = \frac{Xc}{bx}$$

- ✓ Where ' Xc ' is the distance between point 'c' (63%) and ' x ' concentration.
- ✓ ' bx ' is the distance between ' x ' concentration and point 'b' (11%).



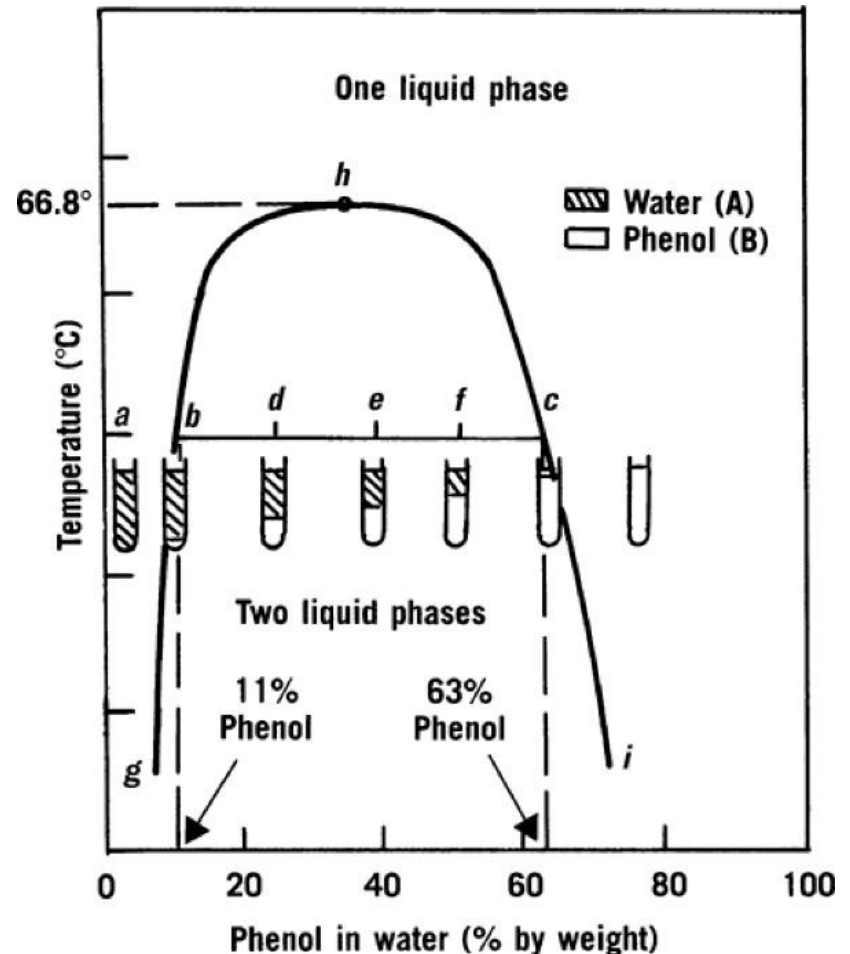
Two-component systems containing liquid phases

Example 1: If we prepare a system containing 24% by weight of phenol and 76% by weight of water (point d), at equilibrium we have two liquid phases present in the test tube.



Example 1....

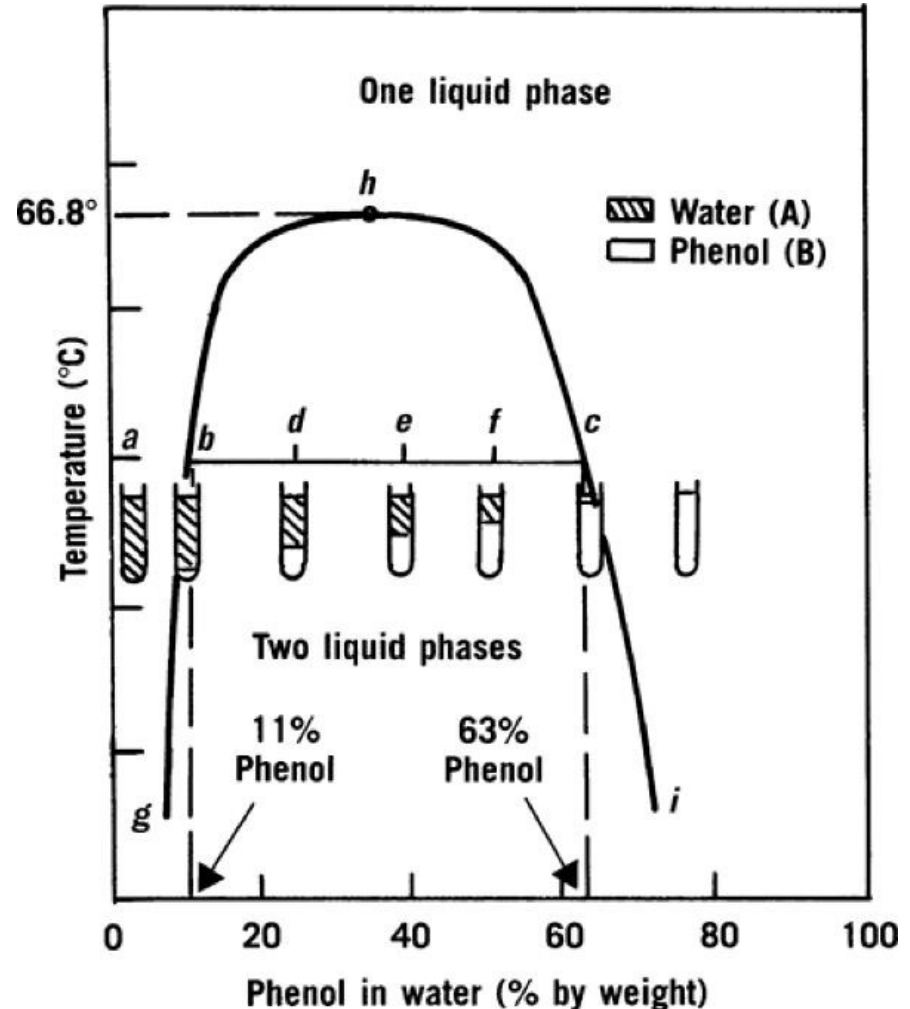
- The upper one, A, has a composition of 11% phenol in water (point **b** on the diagram), whereas the lower layer, B, contains 63% phenol (point **c** on the diagram).



Example 1....

- Phase B will lie below phase A because it is rich in phenol, and phenol has a higher density than water.
- In terms of the relative weights of the two phases:

$$\frac{\text{Weight of phase A}}{\text{Weight of phase B}} = \frac{\text{Length } dc}{\text{Length } bd}$$



Example 1....

- Point b = 11%
- Point c = 63%
- Point d = 24%

$$\text{Wt. A/Wt. B} = (63\% - 24\%) / (24\% - 11\%) = 3/1$$

➔ If there is 100 g of phenol-water system, at equilibrium weight of water phase will be 75 g and weight of phenol phase will be 25 g.

Example 2

- ✓ If 50 g of phenol is used for preparing 50% phenol-water system at 50 °C (point *f* in the diagram), after attaining equilibrium calculate:
 - i. weight of water phase
 - ii. weight of phenol phase
 - iii. weight of phenol and water in water phase
 - iv. weight of phenol and water in phenol phase

Application of two component phase diagram

- The phase diagram is used in practice to formulate systems containing more than one component where it may be advantageous to achieve a single phase product.
- For example, the handling of solid phenol, a necrotic agent, is facilitated in pharmacy if a solution of phenol with water is formulated.

Application of two component phase diagram...

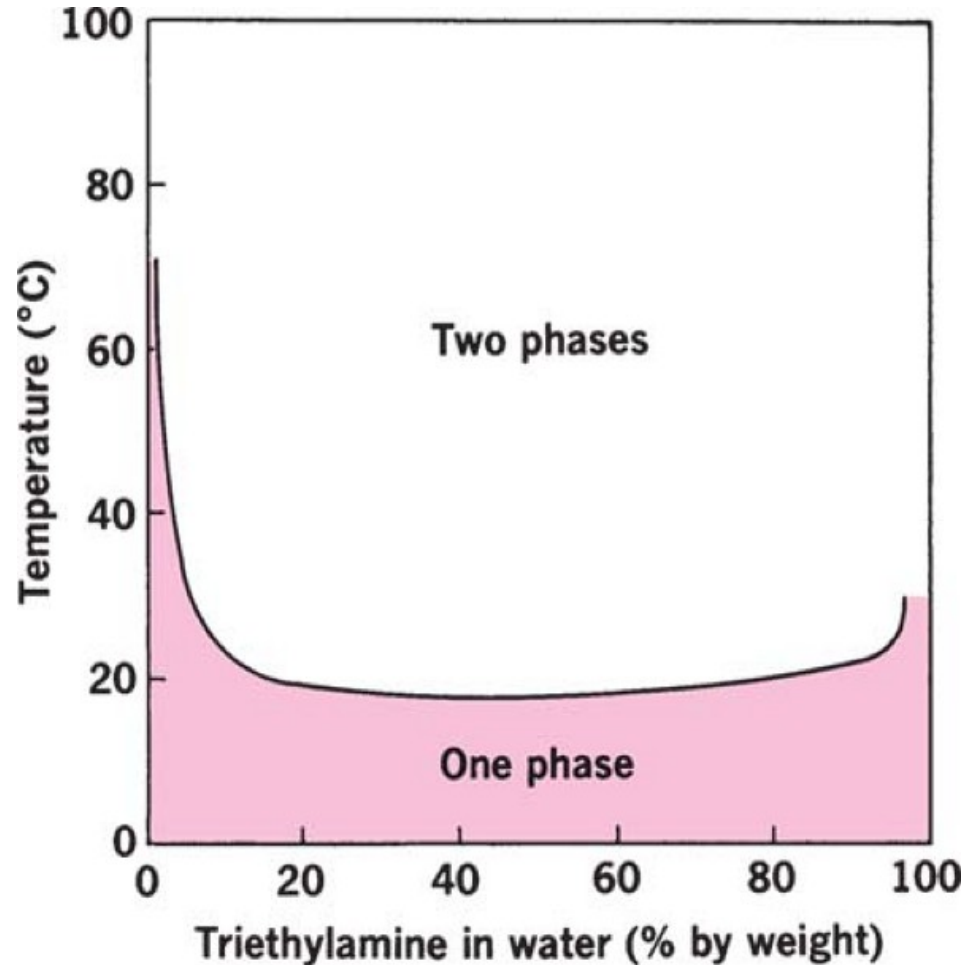
- A number of solutions containing different concentrations of phenol are official in several pharmacopeias.
 - For example, liquified phenol USP, which contains 90% w/w of phenol, freezes at about 17 °C.
 - Problem of inaccuracy during measurement.

Application of two component phase diagram...

- In order to correct this problem, by studying, phenol-water phase diagram, Mulley suggested that the most convenient formulation of a single liquid phase solution was about 76% w/w.
 - This mixture has a freezing point of about 3.5 °C .
 - It is preferred over liquified phenol USP.

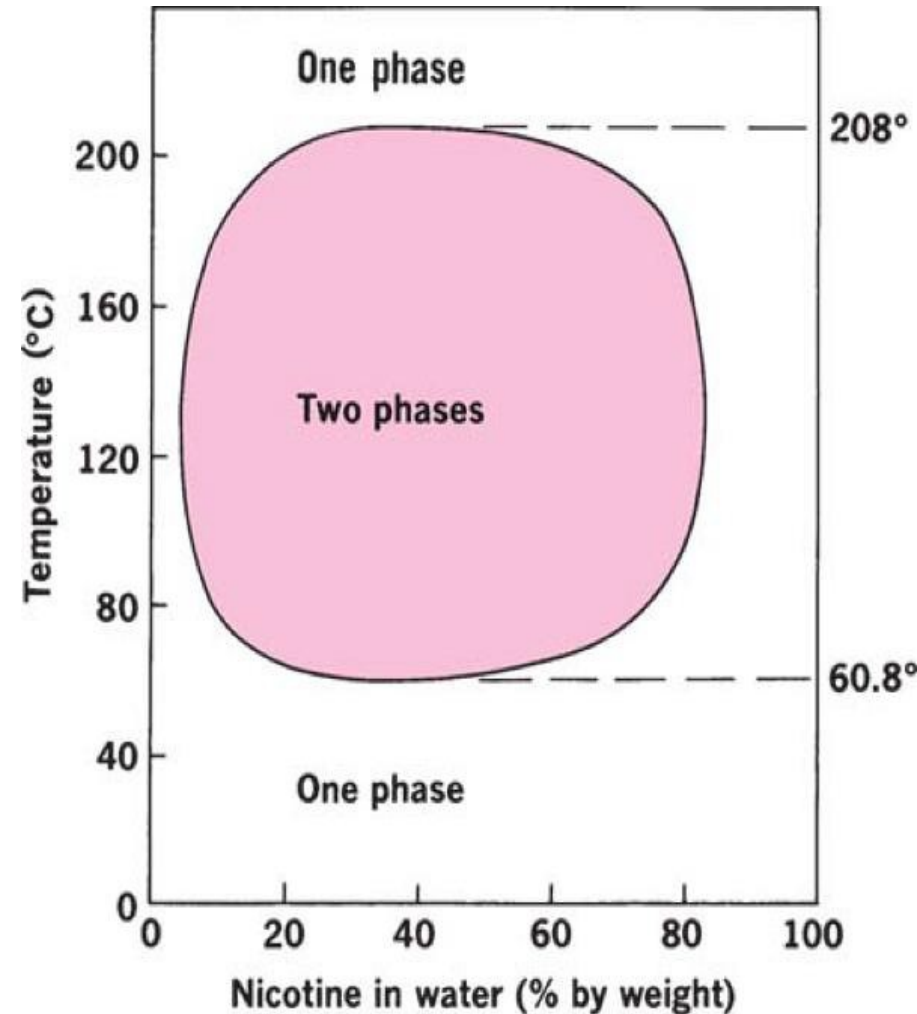
Phase diagram of triethylamine–water system

- Shows *lower consolute temperature*.



Phase diagram of Nicotine–water system

- Shows upper and lower consolute temperatures.



Three component systems (ternary system)

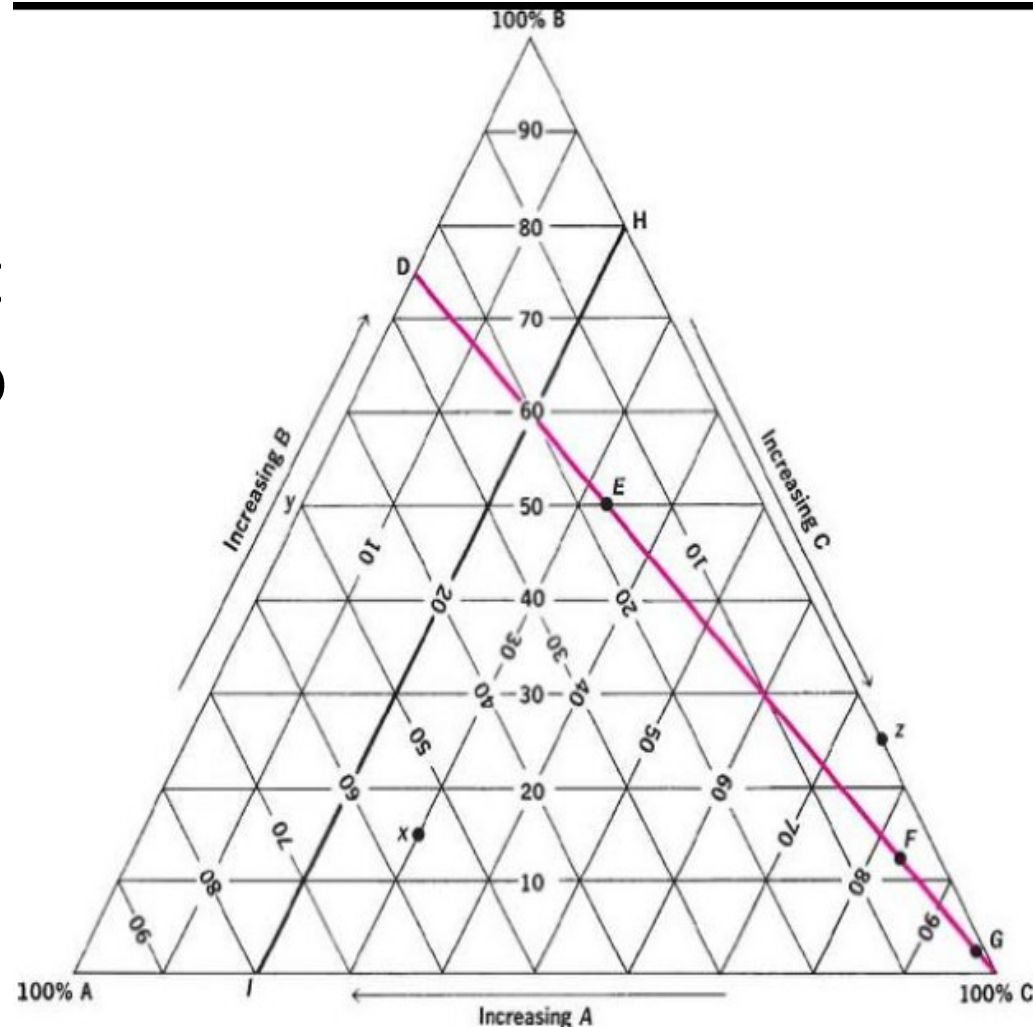
- In systems containing three components but only one phase, $F = 3 - 1 + 2 = 4$ for a non-condensed system.
- The four degrees of freedom are **temperature**, **pressure**, and the **concentrations of two of the three components**.

Three component systems (ternary system)

- Only two concentration terms are required because the sum of the two subtracted from the total will give the concentration of the third component.
- If we regard the system as condensed and hold the temperature constant, then **F becomes 2.**

Three component systems (ternary system)

- Because we are dealing with a three-component system, it is more convenient to use **triangular coordinate graphs**.

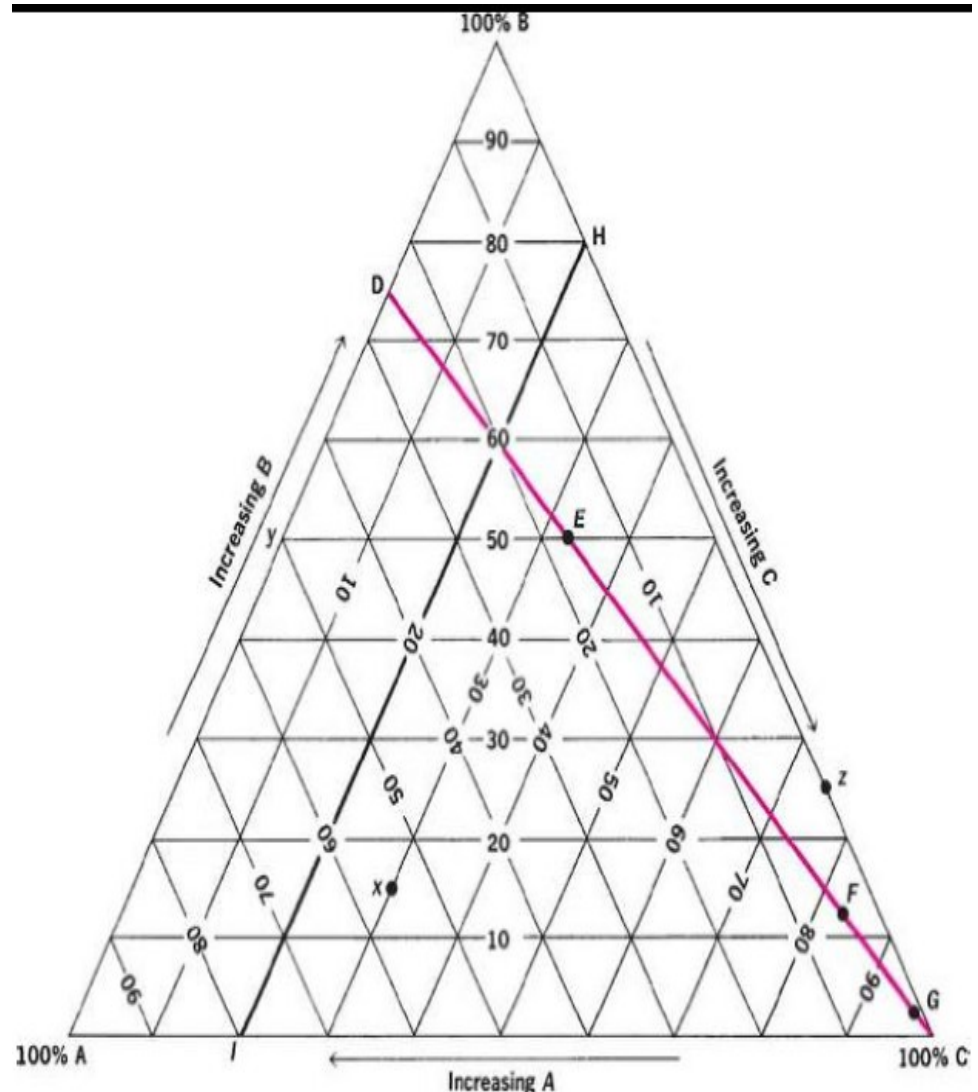


Rules relating to triangular diagrams

- The concentrations in ternary systems are expressed on a weight basis.
- ✓ Each of the three corners or apexes of the triangle represent 100% by weight of one component (A, B, or C). That same apex will represent 0% of the other two components.
- ✓ Each side represents a binary mixture and the interior represents all ternary compositions.

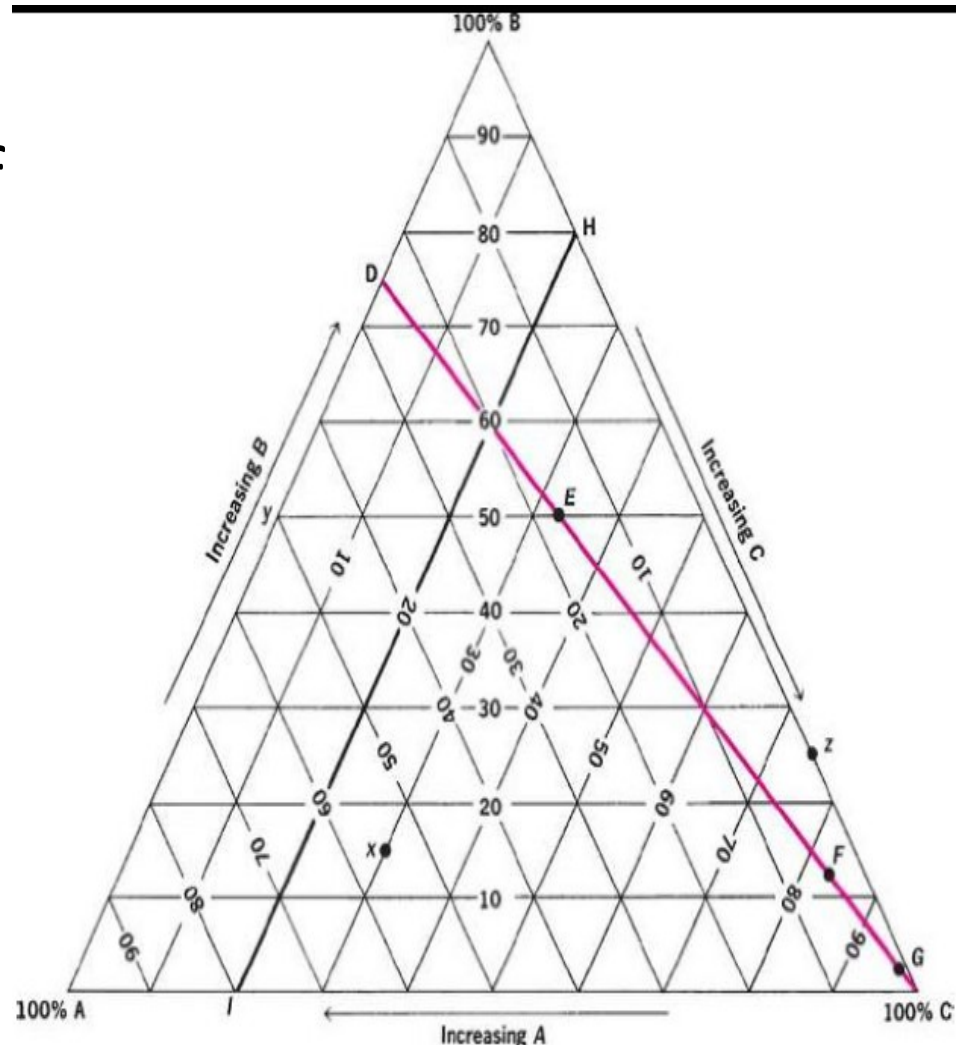
Rules relating to triangular diagrams

- ✓ As one moves along the bounding lines (conventionally in clockwise direction), the concentration of one component increases and the concentration of the other component decreases.



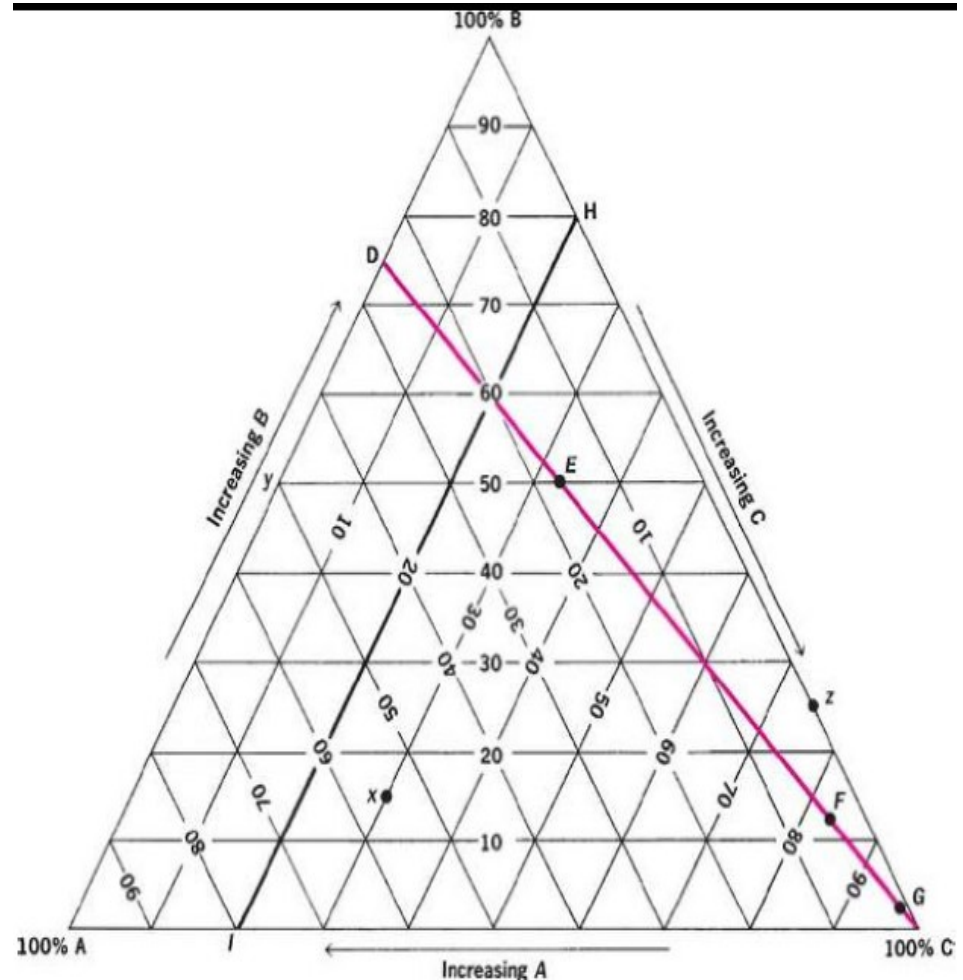
Rules relating to triangular diagrams...

- ✓ Any line drawn parallel to one side of the triangle (for example, line **HI**) represents ternary system in which the **percent by weight of one component is constant**.



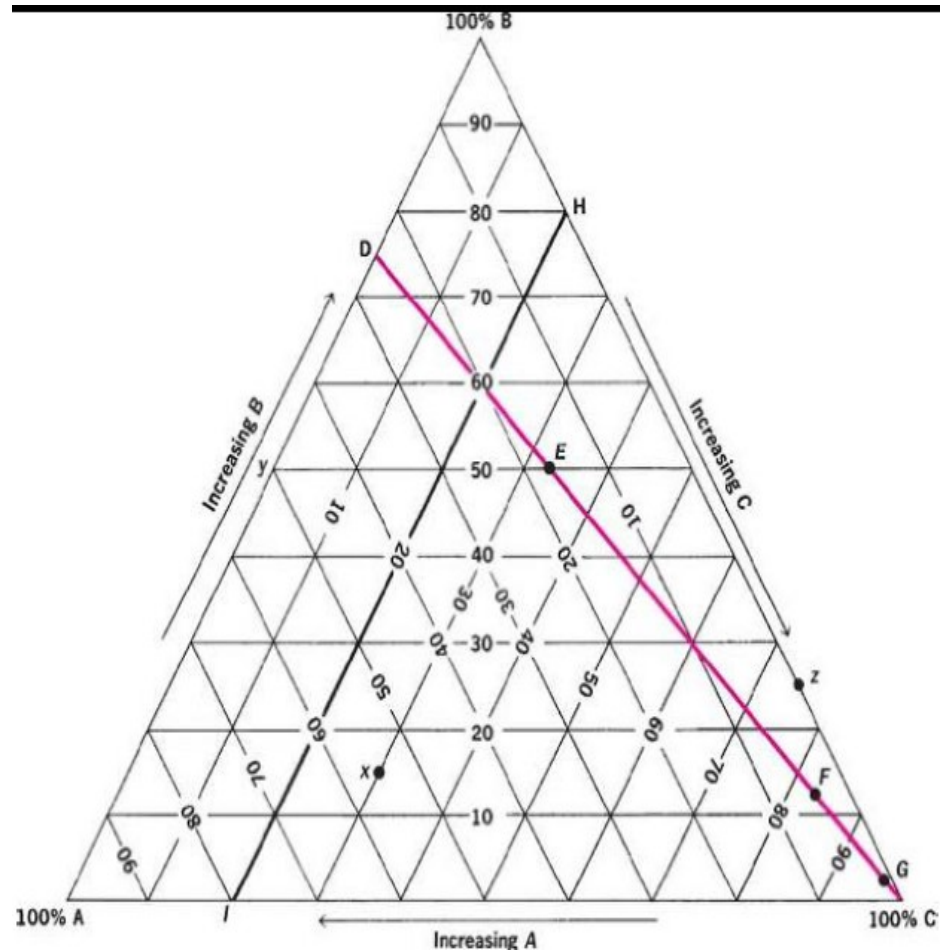
Rules relating to triangular diagrams...

- ✓ For example, all systems prepared along **HI** will contain 20% of C and varying concentrations of A and B.



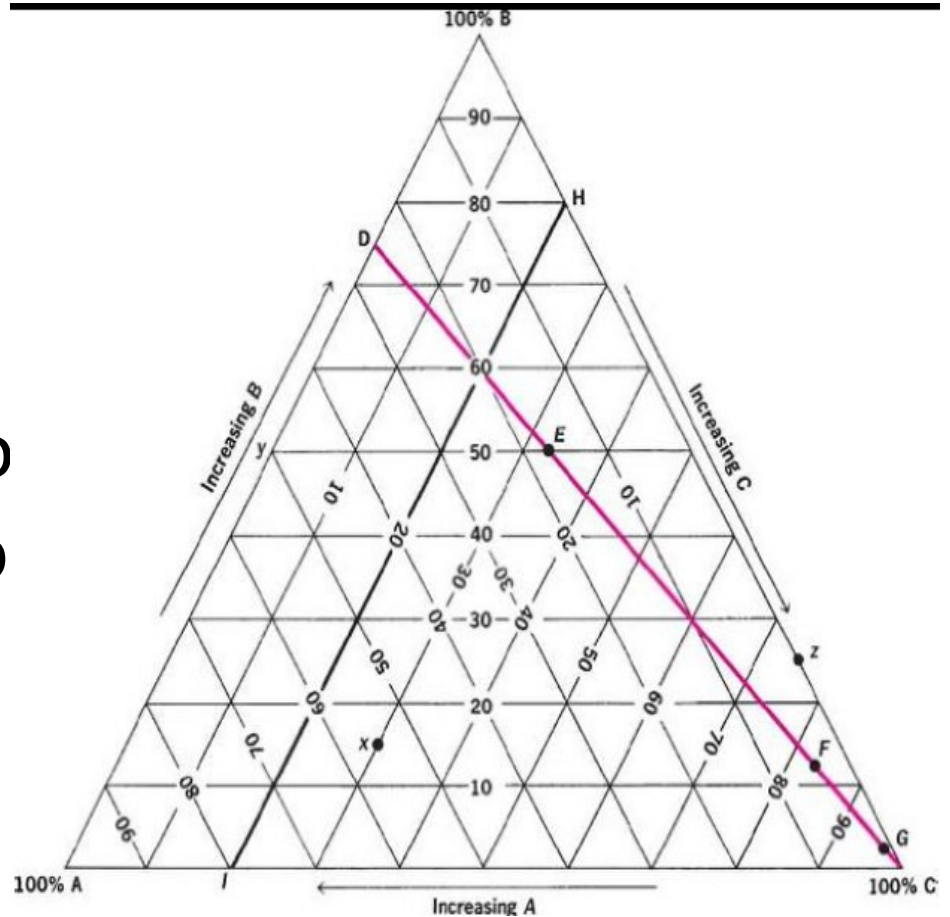
Rules relating to triangular diagrams...

- ✓ If a line is drawn through any apex to a point on the opposite side (e.g., line **DC** in the figure), then all systems represented by points on such a line have a **constant ratio of two components**, in this case A and B.



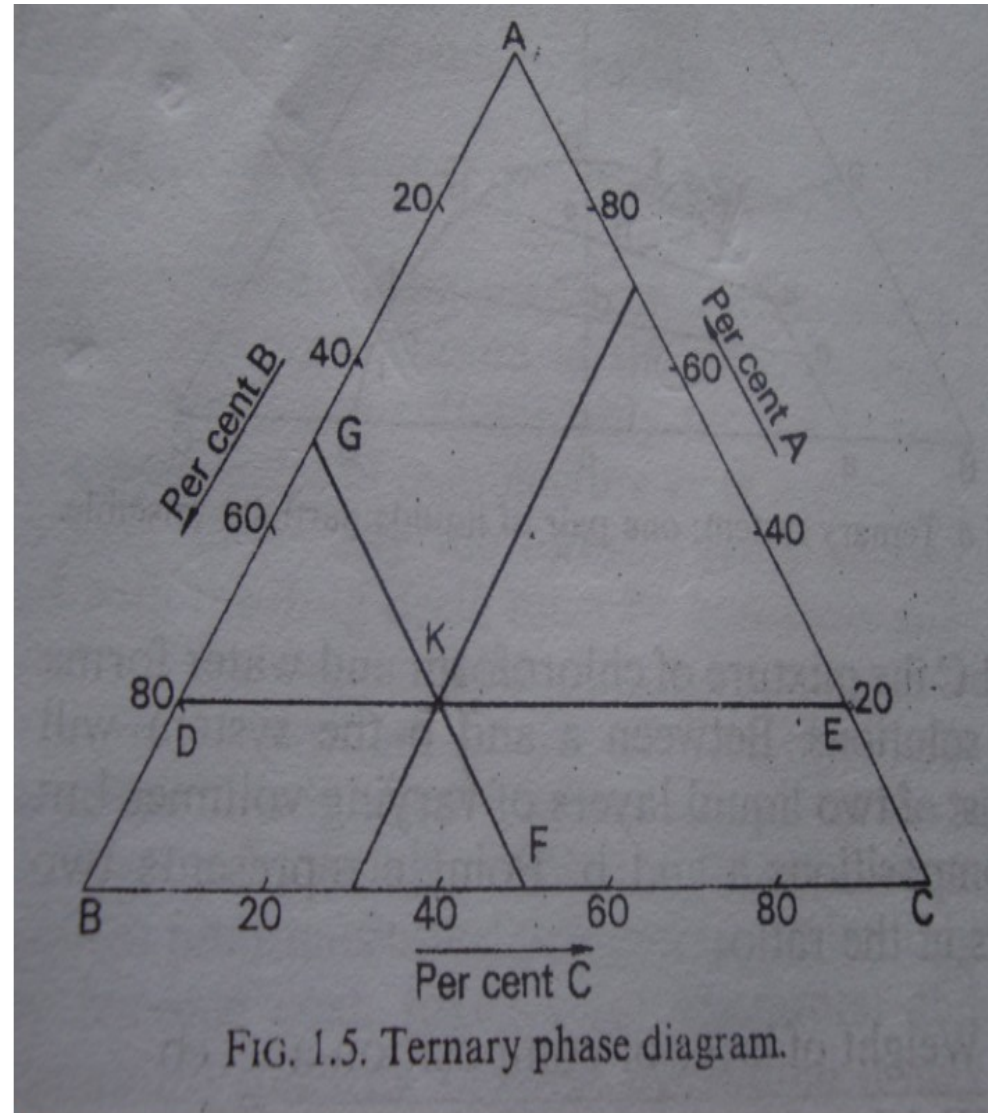
Rules relating to triangular diagrams...

- ✓ The sum of the distances from any given point to the three sides, along the perpendicular to the sides, is equal to the height of the triangle.



Rules relating to triangular diagrams...

- ✓ Example, in the ternary diagram right, at point K, the concentrations of A, B, and C are 20%, 50%, 30%, respectively.



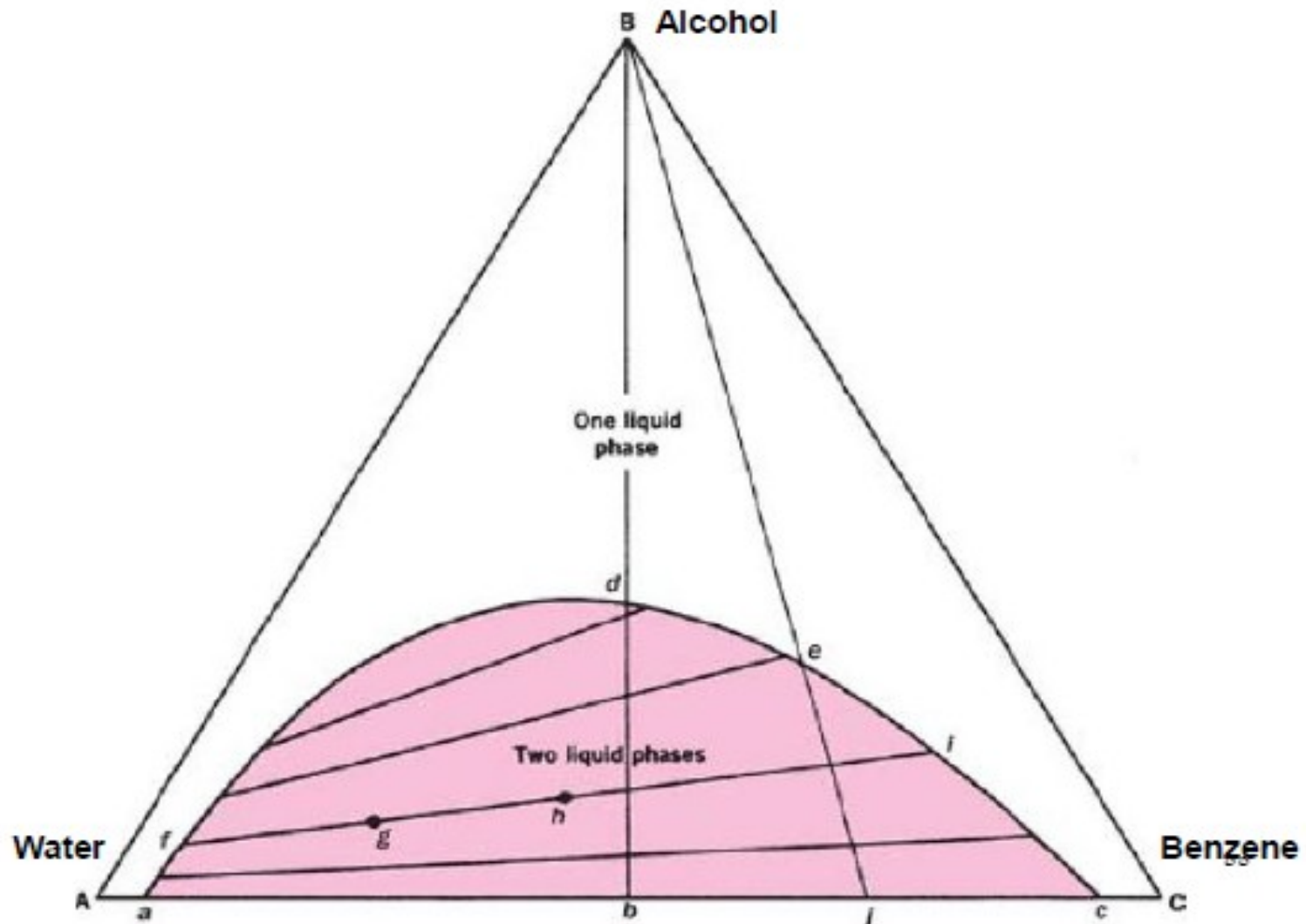
Ternary systems with one pair of partially miscible liquids

- Water and benzene are miscible only to a slight extent, a mixture of the two usually produces a two-phase system.
- The heavier of the two phases consists of water saturated with benzene, while the lighter phase is benzene saturated with water.

Ternary systems with one pair of partially miscible liquids.....

- On the other hand, alcohol is completely miscible with both benzene and water.
- The addition of sufficient alcohol to a two-phase system of benzene and water would produce a single liquid phase in which all three components are miscible.

Ternary diagram of water-alcohol-benzene system



Ternary systems with one pair of partially miscible liquids.....

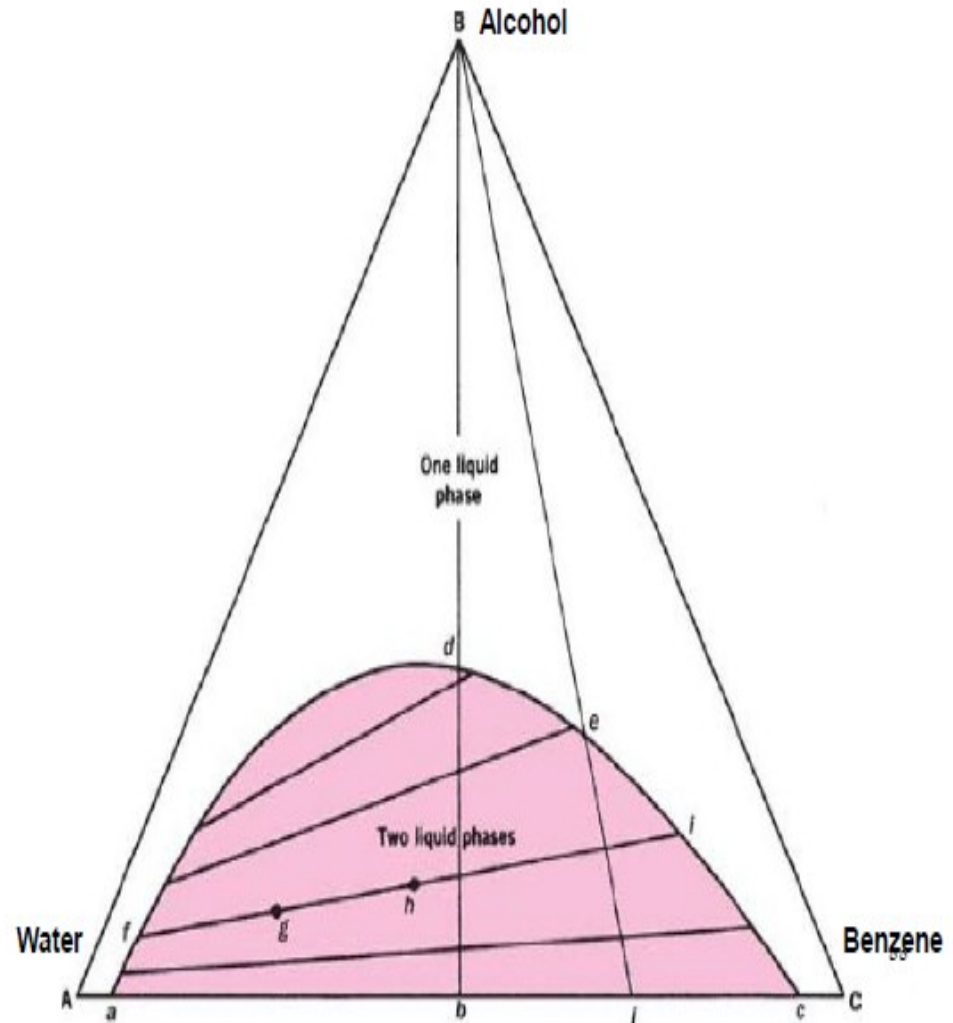
- The curve *afdeic*, (*binodal curve* or *binodal*), marks the extent of the two-phase region.
- The remainder of the triangle contains one liquid phase.
- The tie lines within the binodal are not necessarily parallel to one another or to the base line, AC, as was the case in the two-phase region of binary systems.

Ternary systems with one pair of partially miscible liquids.....

- The directions of the tie lines are related to the shape of the binodal, which in turn depends on the relative solubility of the third component (in this case, alcohol) in the other two components.
- Only when the added component acts equally on the other two components to bring them into solution will the binodal be perfectly symmetric and the tie lines run parallel to the baseline.

Ternary systems with one pair of partially miscible liquids....

- The properties of tie lines discussed for two component system (phenol-water) still applies.
- For example, systems *g* and *h* along the tie line *fi*, both give rise to two phases having the compositions denoted by the points *f* and *i*.



Ternary systems with one pair of partially miscible liquids.....

- The relative amounts, by weight, of the two conjugate phases will depend on the position of the original system along the tie line.
- For example, system *g*, after reaching equilibrium, will separate into two phases, *f* and *i*, hence,

$$\frac{\text{Weight of benzene phase}}{\text{Weight of water phase}} = \frac{\text{distance } fg}{\text{distance } gi}$$